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SQL TASK

1. Employees Above Their Department Average

Display emp_id, emp_name, dept_id, and salary for employees whose salary is greater than the average salary of their own department.

Answer:

```
SELECT emp_id, emp_name, dept_id, salary
FROM Employees e
WHERE salary > (
    SELECT AVG(salary)
    FROM Employees
    WHERE dept_id = e.dept_id
);
```

2. Highest Paid Employee(s) in Every Department

Display the highest-paid employee or employees from each department. If multiple employees in the same department share the highest salary, return all of them.

Answer:

```
SELECT emp_id, emp_name, dept_id, salary
FROM Employees e
WHERE salary = (
    SELECT MAX(salary)
    FROM Employees
    WHERE dept_id = e.dept_id
);
```

3. Second Highest Distinct Salary

Find the second-highest distinct salary in the company without using LIMIT, OFFSET, or JOIN.

Answer:

```
SELECT MAX(salary) AS second_highest_salary
FROM Employees
WHERE salary < (
    SELECT MAX(salary)
    FROM Employees
);
```

4. Employees Receiving the Second Highest Distinct Salary

Display the employee name and salary of every employee whose salary equals the second-highest distinct salary in the company.

Answer:

```
SELECT emp_name, salary
FROM Employees
WHERE salary = (
    SELECT MAX(salary)
    FROM Employees
    WHERE salary < (
        SELECT MAX(salary)
        FROM Employees
    )
);
```

5. Employees Earning More Than Their Manager

Display employees whose salary is greater than the salary of their own manager. Employees without a manager should not appear.

Answer:

```
SELECT emp_id, emp_name, salary, manager_id
FROM Employees e
WHERE manager_id IS NOT NULL
AND salary > (
    SELECT salary
    FROM Employees m
    WHERE m.emp_id = e.manager_id
);
```

6. Managers Who Earn Less Than At Least One Direct Report

Display employees who are managers and whose salary is less than at least one employee reporting directly to them.

Answer:

```
SELECT emp_id, emp_name, salary
FROM Employees e
WHERE EXISTS (
    SELECT 1
    FROM Employees r
    WHERE r.manager_id = e.emp_id
    AND r.salary > e.salary
);
```

7. Departments Whose Average Salary Is Above Company Average

Display the department IDs whose average employee salary is greater than the overall average salary of all employees.

Answer:

```
SELECT dept_id
FROM Employees
GROUP BY dept_id
HAVING AVG(salary) > (
    SELECT AVG(salary)
    FROM Employees
);
```

8. Employees in Departments Whose Average Salary Is Above Company Average

Display all employees who belong to departments whose departmental average salary is greater than the overall company average salary.

Answer:

```
SELECT emp_id, emp_name, dept_id, salary
FROM Employees
WHERE dept_id IN (
    SELECT dept_id
    FROM Employees
    GROUP BY dept_id
    HAVING AVG(salary) > (
        SELECT AVG(salary)
        FROM Employees
    )
);
```

9. Employees Earning More Than Every Employee in HR

Display employees whose salary is greater than the salary of every employee in the HR department. The department ID must be obtained using a subquery from the Departments table.

Answer:

```
SELECT emp_id, emp_name, salary
FROM Employees
WHERE salary > ALL (
    SELECT salary
    FROM Employees
    WHERE dept_id = (
        SELECT dept_id
        FROM Departments
        WHERE dept_name = 'HR'
    )
);
```

10. Employees Earning More Than At Least One Research Employee

Display employees whose salary is greater than at least one employee working in the Research department. Obtain the Research department ID through a subquery.

Answer:

```
SELECT emp_id, emp_name, salary
FROM Employees
WHERE salary > ANY (
    SELECT salary
    FROM Employees
    WHERE dept_id = (
        SELECT dept_id
        FROM Departments
        WHERE dept_name = 'Research'
    )
);
```

11. Departments Having No Projects

Display dept_id and dept_name for departments that do not have any project listed in the projects table.

Answer:

```
SELECT d.dept_id, d.dept_name
FROM Departments d
WHERE NOT EXISTS (
    SELECT 1
    FROM Projects p
    WHERE p.dept_id = d.dept_id
);
```

12. Employees Belonging to Departments With More Than One Project

Display all employees who belong to a department that has more than one project.

Answer:

```
SELECT emp_id, emp_name, dept_id, salary
FROM Employees
WHERE dept_id IN (
    SELECT dept_id
    FROM Projects
    GROUP BY dept_id
    HAVING COUNT(*) > 1
);
```

13. Employees in the Department of the Highest-Budget Project

Display all employees who work in the department that owns the project having the maximum budget. If multiple projects tie for maximum budget, employees from all matching departments should be returned.

Answer:

```
SELECT emp_id, emp_name, dept_id, salary
FROM Employees
WHERE dept_id IN (
    SELECT dept_id
    FROM Projects
    WHERE budget = (
        SELECT MAX(budget)
```

```
        FROM Projects
    )
);
```

14. Employee(s) With Salary Closest to Company Average

Display the employee or employees whose salary has the smallest absolute difference from the company average salary. Do not use ORDER BY ... LIMIT and do not use JOIN.

Answer:

```
SELECT emp_id, emp_name, salary
FROM Employees
WHERE ABS(salary - (
    SELECT AVG(salary)
    FROM Employees
)) = (
    SELECT MIN(
        ABS(salary - (
            SELECT AVG(salary)
            FROM Employees
        ))
    )
    FROM Employees
);
```

15. Departments Whose Top Salary Exceeds Every Manager Salary in That Department

Display department IDs where the maximum employee salary in that department is greater than the salary of every employee in the same department who is acting as a manager of at least one other employee.

Answer:

```
SELECT e.dept_id
FROM Employees e
GROUP BY e.dept_id
HAVING MAX(e.salary) > ALL (
    SELECT m.salary
    FROM Employees m
    WHERE m.dept_id = e.dept_id
    AND EXISTS (
        SELECT 1
        FROM Employees r
        WHERE r.manager_id = m.emp_id
    )
);
```